

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF PHYSICS

FINAL EXAMINATION

PHYS3114 – Electrodynamics

Term 3, 2022

1. Time allowed – 2 hours {+ 15 min for download/reading + 30 min for capture/upload}
2. Total number of questions – 2
3. Total marks available – 30 (pro rata to 60% of course assessment)
4. You are allowed to use your textbook and notes during the exam.
5. Prepare a single file (pdf files preferred) with all of your responses to upload to Moodle by the end of the allotted exam time.
6. Additional time will be allocated for submitting your answers
7. Your submitted file should have your name and zID on the first page of your submitted file (preferably in the header or footer on every page).
8. When uploading please use the following naming convention: PHYS3114 zID SURNAME NAME (e.g., PHYS3114 z5556789 Stark Anthony). NOTE: failure to name your exam file correctly may result in marking delays if your file is misplaced.
9. **Candidates must verify the upload is successful and contact the exam convener immediately if there is an issue.**
10. Answer all questions concisely and legibly.
11. In some cases, sketching your answer can be helpful.
12. Questions are not all worth the same marks; questions are worth the marks shown.
13. You may use a university-approved calculator.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (3)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_o}{D}x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D}y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (4)$$

$$\mathcal{F}\left\{\frac{1}{2}\delta(r-a)\right\} = \pi a J_0(2\pi a \rho) \quad (5)$$

$$\mathcal{F}\left\{\prod\left(\frac{r}{2a}\right)\right\} = a \frac{J_1(2\pi a \rho)}{\rho} \quad (6)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (7)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (8)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (9)$$

$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (10)$$

Bessel function nodes

$J_0(x)=0$, $x = 2.405, 5.5201, 8.6537, \dots$

$J_1(x)=0$, $x = 3.833, 7.0156, 10.1735, \dots$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

PHYS3114 — (more) Definitions and Formulae

Gradient	$\nabla f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$ $\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$ $\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{\partial f}{\partial z} \hat{z}$
Laplacian	$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ $\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}$ $\nabla^2 f = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$
Divergence	$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$ $\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$ $\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$
Curl	$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z}$ $\nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \hat{r}$ $+ \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \hat{\theta} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\phi}$ $\nabla \times \mathbf{A} = \left[\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right] \hat{\phi} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right] \hat{z}$
Identities	$\nabla \times (\nabla f) = 0$ $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$
Gradient Theorem	$\int_a^b (\nabla f) \cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a})$
Divergence Theorem	$\oint_S \mathbf{A} \cdot d\mathbf{a} = \int_V \nabla \cdot \mathbf{A} dv$
Stokes' Theorem	$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a}$

Coulomb's Law	$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$
Biot-Savart Law	$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \hat{\mathbf{r}}}{r^2}$
Lorentz Force Law	$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$
Maxwell's Equations	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ $\nabla \cdot \mathbf{B} = 0 \quad \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$
Electric Potential	$\mathbf{E}(\mathbf{r}) = -\nabla \phi(\mathbf{r}) - \frac{\partial \mathbf{A}}{\partial t}$
Vector Potential	$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}}{r} dv$
Charge Conservation	$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$
Current Density	$\mathbf{J} = \rho \mathbf{v} = \rho \mu \mathbf{E} = \sigma \mathbf{E}$
Poynting Vector	$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$
Stored Energy	$W = \frac{1}{2} \sum_i q_i V(\mathbf{r}_i) = \frac{1}{2} \int V(\mathbf{r}) \rho(\mathbf{r}) d^3\mathbf{r}$ $W = \frac{1}{2} \int \left(\epsilon E^2 + \frac{B^2}{\mu} \right) dv \quad W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} dv$
Faraday's Law	$\int \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi}{dt} \quad \phi = \int_S \mathbf{B} \cdot d\mathbf{a}$
Electric Displacement	$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad \int_S \mathbf{D} \cdot d\mathbf{s} = Q_f$
Magnetic Field Intensity	$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad \int_L \mathbf{H} \times d\mathbf{l} = I_f$
Bound Charge Densities	$\rho_b = -\nabla \cdot \mathbf{P} \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$
Magnetization Current Densities	$\mathbf{J}_b = \nabla \times \mathbf{M} \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$
Linear Media	$\mathbf{P} = \chi_e \epsilon_0 \mathbf{E} \quad \mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}$ $\mathbf{M} = \chi_m \mathbf{H} \quad \mathbf{B} = \mu_0 (1 + \chi_m) \mathbf{H} = \mu \mathbf{H}$
Capacitance	$C = \frac{Q}{V} \quad W = \frac{1}{2} CV^2$
Inductance	$\varepsilon = -L \frac{dI}{dt} \quad W = \frac{1}{2} LI^2$

Physical Constants

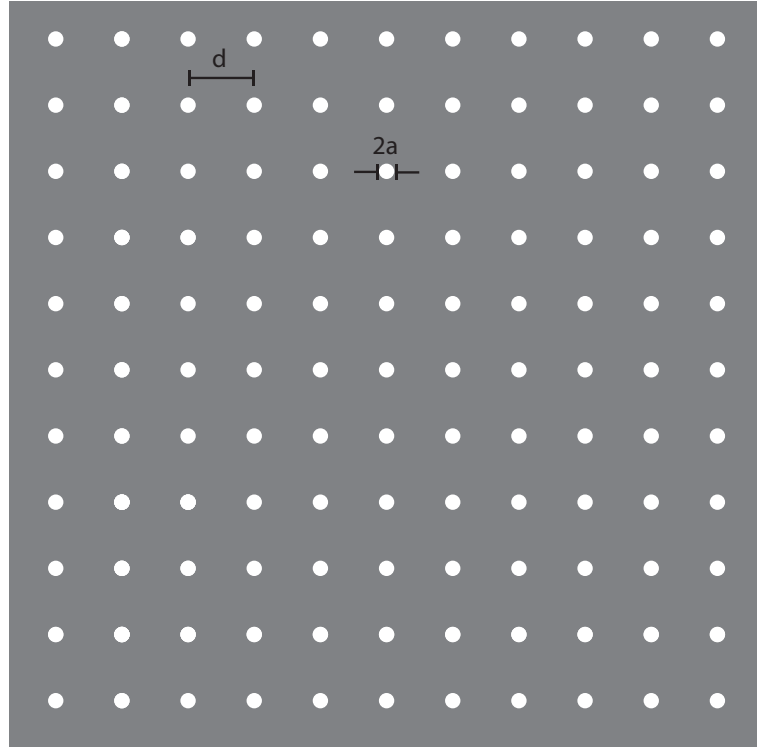
Avagadro's Number	N_A	$6.02 \times 10^{23} \text{ mole}^{-1}$
Permittivity of free space	ϵ_0	$8.85 \times 10^{-12} \text{ Fm}^{-1}$
	$\frac{1}{4\pi\epsilon_0}$	$9.0 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
Permeability of free space	μ_0	$4\pi \times 10^{-7} \text{ Hm}^{-1}$
Electron charge	e	$1.6 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$
Speed of light in vacuum	c	$3.00 \times 10^8 \text{ ms}^{-1}$

Useful Mathematical Values

Bessel function nodes	$J_0(x) = 0$	$x = 2.405, 5.5201, 8.6537, \dots$
Bessel function nodes	$J_1(x) = 0$	$x = 3.833, 7.0156, 10.1735, \dots$

Question 1 (15 marks)

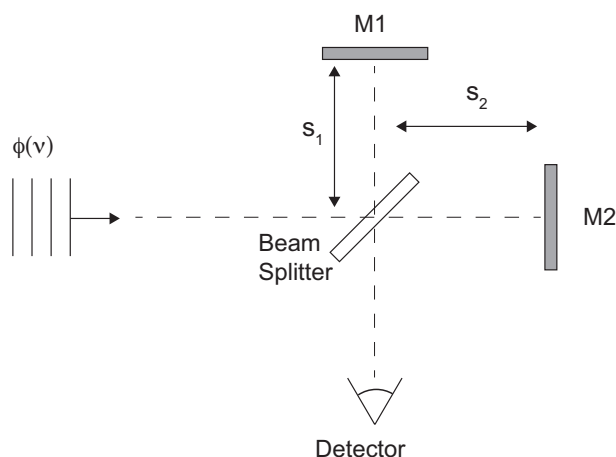
A diffraction aperture consisting of a two-dimensional square array of circular holes is used in a standard far-field diffraction experiment. The diameter of the holes is $2a$ and the periodicity of the square array is d . The circular hole size is small compared to the array spacing and the total number of periods in each dimension is 11 (see Figure below).



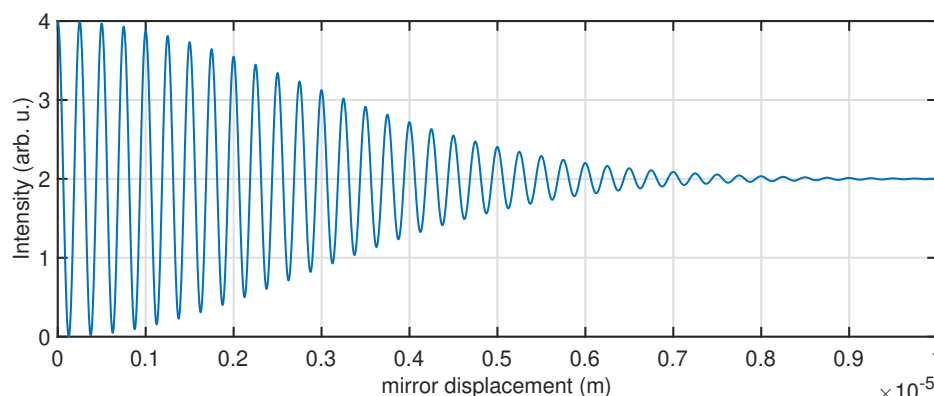
- (a) Briefly describe the conditions for which far-field diffraction is observed.
- (b) Write an expression for the corresponding far-field diffracted field in terms of Fourier variables (u, v) .
- (c) If one single circular hole in the array is blocked (e.g. at location n_x, m_y), write an expression for any additional terms in the diffracted field as a result.
- (d) By considering the mathematical form of the diffracted intensity pattern, explain whether it is possible to distinguish which hole in the array has been blocked.

Question 2 (15 marks)

An interferometer experiment is conducted on light sources of different spectral properties. A diagram of the optical arrangement is provided in the figure below. This is a Michelson type interferometer where the incoming light is split by a partially reflecting mirror (beam splitter) and directed towards either a fixed mirror (M1) or a movable mirror (M2), before being recombined at the detector. The detector at the output measures a time averaged signal $\langle I \rangle_t$. Note that for this problem you may ignore the effects of polarisation.



- Derive an expression for the time averaged intensity of an incident field measured at the detector as a function of the displacement of the movable mirror, s_2 . (hint: delay time)
- An incident light source consists of a delta-function like peak positioned at ν_1 . Using the above result, derive an expression for the time-averaged intensity measured by the detector as a function of mirror displacement for this source.
- A source of unknown spectral properties produces the following response curve corresponding to the time averaged intensity of the detector as a function of relative optical path difference of the interferometer (see figure below). Use the relevant features of the plot to infer the spectral properties of the light source. Marks are awarded for quantitative analysis.



– End of Exam –